Operations Research, Central Exercise 1

LP-Modelling and Graphical Solution

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April 17, 2026

Orga

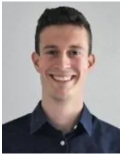
- Central Exercises will be held weekly on Fridays starting at 4.15 p.m.
- TAs will change during the semester according to the covered topics.
- Yes, it is relevant for exams.

(a) Abheek

(b) Halil

(c) Teodora

(d) Jan



(e) Janik

A crash course in convex optimization

$$\max f(x) \text{ s.t. } x \in \mathcal{X} \subseteq \mathbb{R}^n$$

$f : \mathbb{R}^n \to \mathbb{R}$ is concave and $\mathcal{X}$ is convex.

Key geometric objects to understand:

- Tangent cone $T_{\mathcal{X}}(x) = cl(\{d \in \mathbb{R}^n | x + td \in \mathcal{X}, t > 0\})$
- Normal cone $N_{\mathcal{X}}(x) = \{v \in \mathbb{R}^n | \langle v, d \rangle \leq 0 \ \forall d \in T_{\mathcal{X}}(x)\}$.

A point $x^* \in \mathcal{X}$ is optimal if $\nabla f(x^*) \in N_{\mathcal{X}}(x^*)$.

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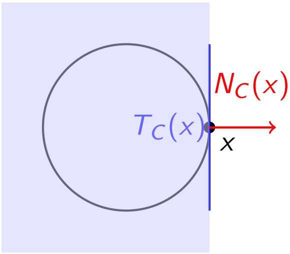
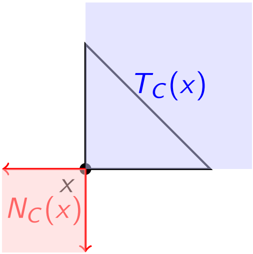


Convex smooth set



Polyhedral set

Linear Program

A special case of convex optimization.

$$\max f(x) \text{ s.t. } x \in \mathcal{X} \subseteq \mathbb{R}^n$$

$$\max c^\top x \text{ s.t. } Ax \leq b, x \geq 0$$

For polytopes of the above form, the normal cone is given by

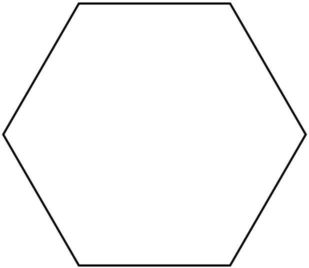
$$\begin{aligned} N_{\mathcal{X}}(x) &= \{A^\top \lambda - \mu | \lambda \geq 0, \mu \geq 0, \lambda_i (a_i^\top x - b_i) = 0, \mu_j x_j = 0\} \\ &= \left\{ \sum_{i \in I(x)} \lambda_i a_i - \sum_{j \in J(x)} \mu_j e_j \middle| \lambda_i, \mu_j \geq 0 \right\}, \end{aligned}$$

where $I(x)$ and $J(x)$ denote the set of active constraints at x.
$I(x) := \{i \mid a_i^\top x = b_i\}$, $J(x) := \{j \mid x_j = 0\}$.

Linear Program - Optimality

$$\max c^\top x \text{ s.t. } Ax \leq b, x \geq 0$$

The normal cone $N_{\mathcal{X}}(x)$ is given by the active constraints at x. x^* is optimal if $\nabla f = c \in N_{\mathcal{X}}(x)$, i.e, if c is in the normal cone spanned by the active constraints at x^*.



Definitions

Linear Program:

$$
\begin{array}{rll}
\text{max/min} & c_1x_1 + c_2x_2 + \dots & \text{Objective function} \\
\text{s.t.} & a_{1,1}x_1 + a_{1,2}x_2 + \dots \le b_1 & \\
 & a_{2,1}x_1 + a_{2,2}x_2 + \dots \le b_2 & \text{Restrictions /} \\
 & \vdots & \text{Constraints} \\
 & a_{m,1}x_1 + a_{m,2}x_2 + \dots \le b_m & \\
 & x_1, x_2, \dots \ge 0 & \text{(Decision) Variables}
\end{array}
$$

Definitions

Linear Program:

$$
\begin{aligned}
\max/\min \quad & c_1 x_1 + c_2 x_2 + \dots && \text{Objective function} \\
\text{s.t.} \quad & a_{1,1} x_1 + a_{1,2} x_2 + \dots \leq b_1 \\
& a_{2,1} x_1 + a_{2,2} x_2 + \dots \leq b_2 && \text{Restrictions /} \\
& \vdots && \text{Constraints} \\
& a_{m,1} x_1 + a_{m,2} x_2 + \dots \leq b_m \\
& x_1, x_2, \dots \geq 0 && \text{(Decision) Variables}
\end{aligned}
$$

$$
\begin{aligned}
\max \quad & \mathbf{c}^\top \mathbf{x} && x, c \in \mathbb{R}^n \\
\text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b} && A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m \\
& \mathbf{x} \geq 0
\end{aligned}
$$

Graphical Solutions

- Useful for solving 2D optimization problems.
- Help to build intuition about feasible regions, objective functions, and optimal solutions.
- Useful for educational purposes and for understanding the geometry of linear programming.
- Impractical for real-world problems, typically involving several thousand variables.

Exercise 1: Toy Production

Problem Description: The Toys&Joys GmbH produces wind-up ducks and toy trains made of wood. One wind-up duck can be sold for 27€. The material cost per duck is 10€, the other variable costs are 14€. The selling price for the toy train is 21€, the material cost and other variable costs per toy train are 9€ and 10€, respectively.

Two operations are necessary in the production: Painting and assembling. Painting the individual parts of a duck takes 2 hours, their assembly 1 hour. For the train, painting and assembly each take 1 hour.

Any number of raw materials can be used each week. However, per week a maximum of 100 hours is available for painting, and a maximum of 80 hours is available for assembly. In addition, the demand for wind-up ducks is limited to 40 pieces per week.

Define a linear program whose solution maximizes the weekly profit under the given restrictions. Afterwards, solve this graphically.

Exercise 1: Toy Production

Problem Description Summary:

- Toy manufacturer produces wind-up ducks and toy trains.
- Constraints on production hours and demand.
- Objective: Maximize weekly profit.

The variables:

x_1: number of wind-up ducks produced per week
x_2: number of toy trains produced per week

Exercise 1: Toy Production (Objective)

One wind-up duck can be sold for 27€. The material cost per duck is 10€, the other variable costs are 14€.
The selling price for the toy train is 21€, the material cost and other variable costs per toy train are 9€ and 10€, respectively.

$$\begin{aligned} & \max\ (27 - 10 - 14)x_1 + (21 - 9 - 10)x_2 \\ \Leftrightarrow\ & \max\ 3x_1 + 2x_2 \end{aligned}$$

Exercise 1: Toy Production (Constraints)

Painting the individual parts of a duck takes 2 hours, their assembly 1 hour.
For the train, painting and assembly each take 1 hour.
However, per week a maximum of 100 hours is available for painting, and a maximum of 80 hours is available for assembly.

Exercise 1: Toy Production (Constraints)

Painting the individual parts of a duck takes 2 hours, their assembly 1 hour.
For the train, painting and assembly each take 1 hour.
However, per week a maximum of 100 hours is available for painting, and a maximum of 80 hours is available for assembly.
Constraint 1: Painting Capacity

$$2x_1 + x_2 \leq 100$$

Exercise 1: Toy Production (Constraints)

Painting the individual parts of a duck takes 2 hours, their assembly 1 hour.
For the train, painting and assembly each take 1 hour.
However, per week a maximum of 100 hours is available for painting, and a maximum of 80 hours is available for assembly.

Constraint 1: Painting Capacity

$$2x_1 + x_2 \leq 100$$

Constraint 2: Assembly Capacity

$$x_1 + x_2 \leq 80$$

Exercise 1: Toy Production (Constraints)

Constraint 3: Wind-up Ducks Demand

In addition, the demand for wind-up ducks is limited to 40 pieces per week.

$$x_1 \leq 40$$

Exercise 1: Toy Production (Mathematical Model)

Mathematical Model:

$$\begin{aligned}
\text{Max} \quad & 3x_1 + 2x_2 \\
\text{s.t.} \quad & 2x_1 + x_2 \leq 100 \\
& x_1 + x_2 \leq 80 \\
& x_1 \leq 40 \\
& x_1, x_2 \geq 0
\end{aligned}$$

Exercise 1: Toy Production (Graphical Solution)

How to visualize an inequality in $\mathbb{R}^2$?

$a_i^T x \leq b \Leftrightarrow a_{i,1}x_1 + a_{i,2}x_2 \leq b_i$

1. $a_{i,1} = 0$: horizontal line
2. $a_{i,2} = 0$: vertical line
3. $x_1 \leq \frac{b_i}{a_{i,1}}$ & $x_2 \leq \frac{b_i}{a_{i,2}}$

Exercise 1: Toy Production (Feasible region)

Painting:

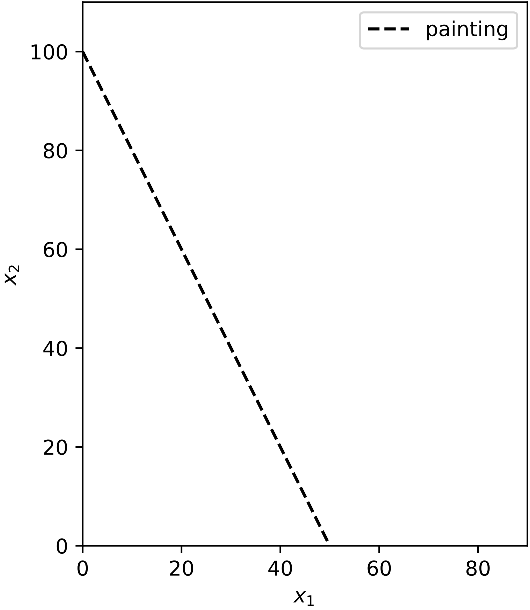
$2x_1 + x_2 \leq 100$

Assembly:

$x_1 + x_2 \leq 80$

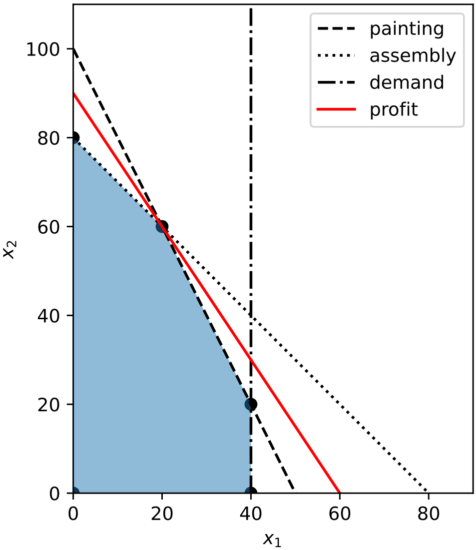
Demand:

$x_1 \leq 40$



Exercise 1: Toy Production (Graphical Solution)

$$
\begin{aligned}
x_1^* &= 20 \\
x_2^* &= 60 \\
z^* &= 3x_1^* + 2x_2^* \\
&= 180
\end{aligned}
$$



Exercise 2: Fuel Oil Trade

Problem Description, Part 1:
In his spare time, Steve trades in fuel oil. For this purpose, he owns a tank with q liters capacity, in which he can temporarily store fuel oil for any length of time. For the coming n days, Steve wants to design a buying and selling plan which maximizes his profit.
On each morning and evening of the n days, a tanker truck with a capacity of r liters arrives at Steve's house. While Steve can only buy (but not sell) fuel oil in the morning, he can only sell (but not buy) fuel oil in the evening. Since the tanker truck also targets other customers, it is already filled with g_i liters of fuel oil in the morning of the i-th day and with h_i liters of fuel oil in the evening of the i-th day, before it arrives at Steve's house.

Exercise 2: Fuel Oil Trade

Problem Description, Part 2:
In addition, the fuel oil price is subject to daily fluctuation. Through extensive research, however, Steve knows that on day i he can buy fuel oil at a price of b_i euros per liter and sell it for s_i euros per liter.
Steve can temporarily store at most as much fuel oil as fits into his tank. Moreover, he cannot sell more fuel oil than he actually owns. Besides, a tanker truck can never take more fuel oil than fits into its tank and can sell at most as much fuel oil as it currently has in its tank.
Set up a linear program whose solution is a buying and selling plan that maximizes Steve's profit. At the beginning of the first day, Steve's tank is already filled with w liters of fuel oil.

Exercise 2: Fuel Oil Trade

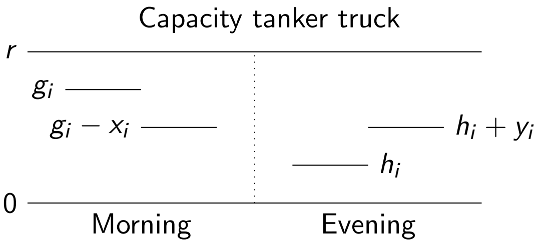
Problem Description Summary:

- Steve trades in fuel oil over n days.
- Constraints on buying and selling, tanker truck capacities.
- Objective: Maximize profit.

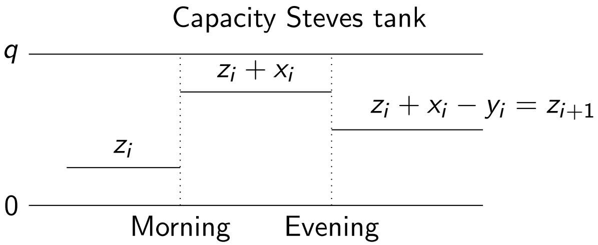
Variables:
$i \in [n]$ the index corresponding to the i-th day
x_i liters of fuel oil that is bought in the morning
y_i liters of fuel oil that is sold in the evening
z_i liters of fuel oil that are in the tank at the beginning of the day.

Fuel Oil Visualization

On each morning and evening of the n days, a tanker truck with a capacity of r liters arrives at Steve's house. While Steve can only buy (but not sell) fuel oil in the morning, he can only sell (but not buy) fuel oil in the evening. Since the tanker truck also targets other customers, it is already filled with g_i liters of fuel oil in the morning of the i-th day and with h_i liters of fuel oil in the evening of the i-th day, before it arrives at Steve's house.



Fuel Oil Visualization



Exercise 2: Fuel Oil Trade (Objective)

Through extensive research, Steve knows that on day i he can buy fuel oil at a price of b_i euros per liter and sell it for s_i euros per liter.
Profit is the difference between sales revenue and purchase costs.

$$\sum_{i=1}^{n} s_i y_i - \sum_{i=1}^{n} b_i x_i$$

Exercise 2: Fuel Oil Trade (Constraints)

Constraint 1: Morning Buying

Since the tanker truck also targets other customers, it is already filled with g_i liters of fuel oil in the morning of the i-th day.

$$x_i \leq g_i \quad \forall i \in [n]$$

Exercise 2: Fuel Oil Trade (Constraints)

Constraint 2: Evening Selling
A tanker truck with a capacity of r liters arrives at Steve's house. He can only sell (but not buy) fuel oil in the evening. Since the tanker truck also targets other customers, it is already filled with h_i liters of fuel oil in the evening of the i-th day, before it arrives at Steve's house.

$$y_i \leq r - h_i \quad \forall i \in [n]$$

Exercise 2: Fuel Oil Trade (Constraints)

Constraint 3: Initial tank

Assume that Steve's tank is already filled with w liters of fuel oil at the beginning of the first day.

$$z_1 = w$$

Exercise 2: Fuel Oil Trade (Constraints)

Constraint 3: Tank level next day

$$z_{i+1} = z_i + x_i - y_i \quad \forall i \in [n-1]$$

Exercise 2: Fuel Oil Trade (Constraints)

Constraint 3: Tank Capacity I
For this purpose, he owns a tank with q liters capacity, in which he can temporarily store fuel oil for any length of time.

$$z_i + x_i \leq q \quad \forall i \in [n]$$

Exercise 2: Fuel Oil Trade (Constraints)

Constraint 3: Tank Capacity II
Steve can only sell oil that he has in his tank.

$$y_i \leq z_i + x_i \quad \forall i \in [n]$$

Exercise 2: Fuel Oil Trade (Mathematical Model)

Mathematical Model:

$$
\begin{array}{ll}
\text{Maximize} & \displaystyle\sum_{i=1}^{n} s_i y_i - \sum_{i=1}^{n} b_i x_i \\
\text{subject to} & x_i \le g_i \quad \forall i \in [n] \\
& y_i \le r - h_i \quad \forall i \in [n] \\
& z_1 = w \\
& z_{i+1} = z_i + x_i - y_i \quad \forall i \in [n-1] \\
& z_i + x_i \le q \quad \forall i \in [n] \\
& y_i \le z_i + x_i \quad \forall i \in [n] \\
& x_i, y_i \ge 0 \quad \forall i \in [n]
\end{array}
$$

Exercise 2: Fuel Oil Trade (Mathematical Model)

Constraints involving z:

$$\begin{aligned} z_1 &= w \\ z_{i+1} &= z_i + x_i - y_i \quad \forall i \in [n-1] \\ z_i + x_i &\leq q \quad \forall i \in [n] \\ y_i &\leq z_i + x_i \quad \forall i \in [n] \end{aligned}$$

Alternative without z:

$$w + \sum_{i=1}^{d} x_i - \sum_{i=1}^{d-1} y_i \leq q \quad \forall d \in [n], \quad (y_0 = -y_1)$$

$$w + \sum_{i=1}^{d} x_i - \sum_{i=1}^{d} y_i \geq 0 \quad \forall d \in [n]$$

Exercise 2: Outlook

This model makes the unrealistic assumption that Steve knows the exact fuel prices for the coming days, as well as the precise amount of fuel that will be available in the tanker truck. In reality, these values are uncertain and not known in advance. While forecasts may be available, they are subject to change. As time progresses and predictions become more accurate, Steve would need to adjust his strategy accordingly. In hindsight, he can compare his strategy to the one obtained by the LP.

Exercise 3: Fitness

Problem Description, Part 1:
George has signed up at the gym in order to be fully trained by the swimming season. His goal is to train the muscle groups $g = 1, \ldots, n$ and to reach a training level of T_g in each of them.

George can train up to 1000 minutes until the start of the swimming season. He has the option of either training alone or hiring a personal trainer. He has to pay 1 euro for every minute he is supervised by his personal trainer.

The first 400 minutes that George trains without a trainer cost him nothing, and for each additional minute that he trains alone, he has to pay 0.10 euros.

Exercise 3: Fitness

Problem Description, Part 2:
In the studio m machines $M_1, \ldots, M_m$ are available for George to train with.
If he trains alone on machine M_j, every minute of training contributes a_{jg} to training level T_g;
every minute of training under the guidance of the personal trainer on M_j contributes $p_{jg} \geq a_{jg}$.

Create a training plan for George so that he achieves his desired level of training in all muscle groups in the time available to him, while minimizing his costs.

Exercise 3: Fitness

Variables:

x_j^p	Minutes that George trains on machine M_j under the supervision of the personal trainer
x_j^a	Minutes that George trains alone on machine M_j
x^a	Minutes that George trains alone, which exceed his 400 free minutes

Exercise 3: Fitness (Objective)

George pays 0.10€ for each minute of training alone, which exceeds his 400 free minutes, and 1€ for each minute with his personal trainer.

$$0.1x^a + \sum_{j=1}^{m} x_j^p$$

Exercise 3: Fitness (Constraints)

For each muscle group g, at least the training level T_g must be reached. Depending on whether George trains alone or with a personal trainer on a machine j, the contribution to the training level changes.

$$\sum_{j=1}^{m} \left(a_{jg} x_j^a + p_{jg} x_j^p \right) \geq T_g \quad \forall g \in [n]$$

Exercise 3: Fitness (Constraints)

In total George must not train more than 1000 minutes.

$$\sum_{j=1}^{m} \left(x_j^a + x_j^p \right) \leq 1000$$

Exercise 3: Fitness (Constraints)

George has a maximum of 400 free minutes training alone.

$$x^a \geq \sum_{j=1}^{m} x_j^a - 400$$

Exercise 3: Fitness (Mathematical Model)

$$
\begin{aligned}
\min 0.1x^a + \sum_{j=1}^{m} x_j^p & \\
\sum_{j=1}^{m} \left(a_{jg} x_j^a + p_{jg} x_j^p \right) \geq T_g & \qquad \forall g \in [n] \\
\sum_{j=1}^{m} \left(x_j^a + x_j^p \right) \leq 1000 & \\
x^a \geq \sum_{j=1}^{m} x_j^a - 400 & \\
x_j^a, x_j^p \geq 0 & \qquad \forall j \in [m] \\
x^a \geq 0 &
\end{aligned}
$$

Linear Modelling

Given a non-linear objective or constraint, can we rewrite this in a linear fashion to utilize LP-solvers?

Example

$$\begin{aligned} &\max\{\min\{x_1, x_2\}\} \\ s.t. &|x_1| \leq 2 \\ &x_2 \geq 0 \\ &x_1 + x_2 \leq 2 \end{aligned}$$

Linear Modelling

Given a non-linear objective or constraint, can we rewrite this in a linear fashion to utilize LP-solvers?

Example

$$
\begin{aligned}
&\max \min\{x_1, x_2\} \\
&s.t. |x_1| \leq 2 \\
&\quad x_2 \geq 0 \\
&\quad x_1 + x_2 \leq 2
\end{aligned}
$$

Solution

$$
\begin{aligned}
&\max\ z \\
&s.t. z \leq x_1 \\
&\quad z \leq x_2 \\
&\quad x_1 \leq 2 \\
&\quad -x_1 \leq -2 \\
&\quad x_2 \geq 0 \\
&\quad x_1 + x_2 \leq 2
\end{aligned}
$$

Approach: Given a non-linear function $f(x)$, introduce an artificial variable z and pose constraints on z to ensure $z = f(x)$.

Linear Modelling

This might not always be directly applicable.
Example 1:

$$\max \max_i \{x_i\} \text{ s.t.} Ax \leq b$$

$$\Leftrightarrow \max \sum k_i x_i \text{ s.t.} Ax \leq b, k_i \in \{0,1\}, \sum k_i = 1.$$

$$\Leftrightarrow \max \sum y_i \text{ s.t.} Ax \leq b, k_i \in \{0,1\}, \sum k_i = 1.$$
$$\&\text{linear constraints enforcing } y_i = k_i x_i$$

Example 2: $\max_i x_i \leq c$ (easy, convex set) vs. $\max_i x_i \geq c$ (hard, non-convex set).

Comparison: Standard vs Normal Form of LP

Normal Form of LP:

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Standard Form of LP:

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax = b \quad (b \geq 0) \\ & x \geq 0 \end{aligned}$$

Conversion: Normal Form to Standard Form

Normal Form (Given):

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Steps to Convert:

1. Add slack variables s to convert inequalities to equalities: $Ax + s = b$.
2. Rewrite the LP in terms of x and s.

Example:

$$3x_1 + 2x_2 \leq 4 \Leftrightarrow 3x_1 + 2x_2 + s_1 = 4$$

Conversion: Standard Form to Normal Form

Normal Form (Given):

$$\begin{aligned} \text{Maximize} \quad & c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Steps to Convert:
Split equalities into two inequalities
$Ax = b \Leftrightarrow b \leq Ax \leq b \Leftrightarrow Ax \leq b$ and $-Ax \leq -b$
Example:

$$3x_1 + 2x_2 = 4 \Leftrightarrow 3x_1 + 2x_2 \leq 4 \text{ and } -3x_1 - 2x_2 \leq -4$$